# The Integral Test — Formal Proof

Series and Integral Transforms, Course 21183
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**Theorem 1** (Integral Test)**.** *Let $f : [1, \infty) \to \mathbb{R}$ be a function that is* ***positive****,* ***continuous****, and* ***decreasing*** *on $[1, \infty)$. Define $a_n = f(n)$. Then*

$$\sum_{n=1}^{\infty} a_n \quad \textit{converges} \quad \iff \quad \int_1^{\infty} f(x)\, dx \quad \textit{converges.}$$

**Key Inequality.** Since $f$ is decreasing, for every integer $n \geq 1$ and every $x \in [n,\, n+1]$:

$$f(n+1) \leq f(x) \leq f(n).$$

Integrating over $[n,\, n+1]$ (interval of length 1):

$$f(n+1) \;\leq\; \int_n^{n+1} f(x)\, dx \;\leq\; f(n).$$

Summing from $n = 1$ to $n = N - 1$:

$$\sum_{n=1}^{N-1} f(n+1) \;\leq\; \int_1^N f(x)\, dx \;\leq\; \sum_{n=1}^{N-1} f(n),$$

which gives us the fundamental double inequality:

$$\sum_{n=2}^{N} a_n \;\leq\; \int_1^N f(x)\, dx \;\leq\; \sum_{n=1}^{N-1} a_n. \tag{1}$$

*Proof.* Let $S_N = \sum_{n=1}^{N} a_n$ denote the $N$-th partial sum of the series. Note that $\{S_N\}$ is *increasing* since all terms $a_n = f(n) > 0$.

**($\Leftarrow$) If the integral converges, then the series converges.**
Suppose $\int_1^{\infty} f(x)\, dx = L < \infty$. From the right inequality in (**??**):

$$\int_1^N f(x)\, dx \;\leq\; \sum_{n=1}^{N-1} a_n = S_{N-1} \leq S_N.$$

Using the left inequality in (**??**):

$$S_N = a_1 + \sum_{n=2}^{N} a_n \;\le\; a_1 + \int_1^N f(x)\,dx \;\le\; a_1 + L.$$

So $\{S_N\}$ is increasing and bounded above by $a_1+L$. By the Monotone Convergence Theorem, $\{S_N\}$ converges, i.e. $\sum_{n=1}^{\infty} a_n$ converges.

**($\Rightarrow$) If the series converges, then the integral converges.**

Suppose $\sum_{n=1}^{\infty} a_n = S < \infty$. The function $I(t) = \int_1^t f(x)\,dx$ is increasing in $t$ (since $f > 0$). From the right inequality in (**??**):

$$\int_1^N f(x)\,dx \;\le\; \sum_{n=1}^{N-1} a_n \;\le\; S.$$

So $I(N)$ is increasing and bounded above by $S$. By the Monotone Convergence Theorem, $\lim_{N\to\infty} I(N)$ exists, i.e. $\int_1^{\infty} f(x)\,dx$ converges. $\square$

**Remark.** The two directions are completely symmetric in structure: in both cases we have a monotone increasing quantity (partial sums or integral), and we use inequality (**??**) to bound it above, then apply the Monotone Convergence Theorem.

**Example.** The $p$-series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ with $f(x) = x^{-p}$ (positive, continuous, decreasing for $p > 0$, $x \ge 1$).

$$\int_1^{\infty} \frac{dx}{x^p} = \begin{cases} \dfrac{1}{p-1} & \text{if } p > 1 \quad \text{(converges)}, \\ +\infty & \text{if } p \le 1 \quad \text{(diverges)}. \end{cases}$$

Therefore $\sum 1/n^p$ converges if and only if $p > 1$.